Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY – PUNE
[An autonomous Institute affiliated to Savitribai Phule University]
Bibwewadi and Kondhwa campuses

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Tutorial 3
Demonstrate random sampling & compare sample vs population mean

Code: -

```python
# Experiment: Random Sampling and Comparison of Sample Mean vs Population Mean

# Step 1: Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Step 2: Create Dataset (Population)
np.random.seed(42)

data = {
    "Student_ID": range(1, 101),
    "Exam_Score": np.random.randint(50, 100, 100)
}

df = pd.DataFrame(data)

print("First 5 records of population dataset:")
print(df.head())

# Step 3: Calculate Population Mean
population_mean = df["Exam_Score"].mean()
print("\nPopulation Mean:", population_mean)

# Step 4: Perform Random Sampling
sample = df.sample(n=20)

print("\nRandom Sample (20 students):")
print(sample)
```

```python
# Step 5: Calculate Sample Mean
sample_mean = sample["Exam_Score"].mean()
print("\nSample Mean:", sample_mean)

# Step 6: Compare Population Mean and Sample Mean
print("\nComparison:")
print("Population Mean:", population_mean)
print("Sample Mean:", sample_mean)

# Step 7: Visualize the Comparison
labels = ["Population Mean", "Sample Mean"]
values = [population_mean, sample_mean]

plt.bar(labels, values)
plt.title("Population Mean vs Sample Mean")
plt.ylabel("Mean Score")
plt.show()
```

OUTPUT: -

```
First 5 records of population dataset:
   Student_ID  Exam_Score
0           1          88
1           2          78
2           3          64
3           4          92
4           5          57

Population Mean: 74.07

Random Sample (20 students):
    Student_ID  Exam_Score
92          93          94
78          79          97
23          24          70
49          50          75
81          82          63
16          17          71
53          54          96
15          16          52
79          80          64
18          19          73
17          18          51
9           10          60
20          21          79
45          46          74
68          69          53
67          68          91
86          87          94
91          92          74
37          38          52
66          67          55

Sample Mean: 71.9
```

```
Comparison:
Population Mean: 74.07
Sample Mean: 71.9
```